

数学练习 第 05 份 答案与解析

共 75 题 与《练习05》题号一一对应

1. 齐次线性方程的解全体为 $y = Ce^{-\int P}$, 两特解 $y_1 = C_1 e^{-\int P}$, $y_2 = C_2 e^{-\int P}$.
则 $y_1 - y_2 = (C_1 - C_2)e^{-\int P}$ 不是常数 (因 $P \neq 0$), 故 A 错误; B、C、D 均正确.
2. $AB = O, B \neq O \Rightarrow r(A) + r(B) \leq 3$ 且 $r(A) \leq 2$, 故 $|A| = -(k-1)^2(k+3) = 0$, $k = 1$ 或 -3 .
 $k = 1$ 时 $r(A) = 1$, $r(B)$ 可为 1 或 2, 不能确定;
 $k = -3$ 时 $r(A) = 2$, 故 $1 \leq r(B) \leq 1$, 即 $r(B) = 1$. 选 B.

3. A 反例 $u_n = n, v_n = \frac{1}{n^3}$; C 反例 $u_n = -n, v_n = \frac{1}{n^2}$; D 显然错.
B: 由 $(u_n + v_n)^2 \leq 2(u_n^2 + v_n^2)$ 及比较判别法知收敛. 选 B.

4. 由 $\sum_{k=0}^{\infty} \frac{C}{k!} = Ce = 1$ 得 $C = e^{-1}$, 即 $X \sim P(1)$,

$$E(X^2) = DX + (EX)^2 = 1 + 1 = 2.$$

5. D 关于 y 轴对称而 $\frac{x}{2x^2+y^2}$ 关于 x 为奇函数, 故只需算 $\iint_D \frac{d\sigma}{2x^2+y^2}$.
在右半部分作变换 $u = x, v = y - x$ (雅可比为 1), 区域变为 $1 \leq u^2 + v^2 \leq 2, u \geq 0, v \geq -u$,
被积函数变为 $\frac{1}{3u^2 + 2uv + v^2}$. 用极坐标 ($r \in [1, \sqrt{2}], \varphi \in [-\frac{\pi}{4}, \frac{\pi}{2}]$):

$$I = 2 \int_1^{\sqrt{2}} \frac{dr}{r} \int_{-\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{d\varphi}{2 + \cos 2\varphi + \sin 2\varphi} = 2 \cdot \frac{\ln 2}{2} \cdot \frac{\sqrt{2}\pi}{4} = \frac{\sqrt{2}\pi \ln 2}{4}.$$

6. 由通解知 $r(A) = 1, A(-2, 1)^T = 0, A(3, -4)^T = (3, 2)^T$.
将 $\beta = (5, -10)^T$ 分解: $\beta = 2(-2, 1)^T + 3(3, -4)^T$, 故 $A\beta = 3(3, 2)^T = (9, 6)^T$,
 $\beta^T A\beta = 5 \cdot 9 + (-10) \cdot 6 = -15$.

7. 检验方差: $H_0: \sigma^2 \leq 15^2$.

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} = \frac{9 \times 900}{225} = 36 > \chi_{0.05}^2(9) = 16.919,$$

落入拒绝域, 故认为标准差显著超过 15 g, 机器生产不正常.

8. L 由两条半径段和一段圆弧组成:

$$\int_{\theta=0}^{\theta=\frac{\pi}{4}} = \int_{\theta=\frac{\pi}{4}}^{\theta=\frac{\pi}{2}} = \int_0^a e^r dr = e^a - 1, \quad \int_{r=a}^r = e^a \cdot a \cdot \frac{\pi}{4}.$$

$$\text{故 } I = 2(e^a - 1) + \frac{\pi a}{4} e^a.$$

9. 原式 = $\int_1^{+\infty} \frac{dx}{x\sqrt{x-1}}; \xrightarrow{t=\sqrt{x-1}}; \int_0^{+\infty} \frac{2dt}{1+t^2} = \pi$. 选 C.

10. $\text{Cov}(X, Y) = \text{Cov}(X, X^2 + X) = EX^3 + EX^2 - EX \cdot E(X^2 + X) = 1 \neq 0$,

故相关; 又 Y 是 X 的函数, 必不独立. 选 A.

11. L_1 : 点 $A(-2, 3, -1)$, 方向 $s_1 = (1, -1, 1)$; L_2 : 点 $B(-4, 0, 4)$, 方向 $s_2 = (2, 1, 3)$.
公垂线方向 $s = s_1 \times s_2 = (-4, -1, 3)$. 过 L_1 且含 s 的平面: $2x + 7y + 5z - 12 = 0$;
过 L_2 且含 s 的平面: $3x - 9y + z + 8 = 0$. 故公垂线为

$$\begin{cases} 2x + 7y + 5z - 12 = 0, \\ 3x - 9y + z + 8 = 0, \end{cases} \quad \text{即} \quad \frac{x}{4} = \frac{y-1}{1} = \frac{z-1}{-3}.$$

12. (I) $r(A) = 1$, A 的行向量与 $(-1, 1, 0), (2, 0, 1)$ 正交, 故各行都平行于 $(1, 1, -2)$.
设 $A = \eta(1, 1, -2)$, 由 $A(1, 1, -2)^T = b$ 得 $6\eta = b$, $\eta = (1, 1, -2)^T$, 即

$$A = \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ -2 & -2 & 4 \end{pmatrix} = \eta\eta^T.$$

(II) $A^2 = \eta(\eta^T\eta)\eta^T = 6A$, 取 $\alpha = (1, 1, -2)^T$, 则 $A^2 = \alpha\beta$ 中
 $\beta = (6, 6, -12)$.

(III) $X^TAX = (\eta^TX)^2 = 0 \iff x_1 + x_2 - 2x_3 = 0$, 故全部解为
 $X = k_1(-1, 1, 0)^T + k_2(2, 0, 1)^T$.

13. 令 $u = x - 2$, $y = (u^2 - 1)^2 = u^4 - 2u^2 + 1$, $y'' = 12u^2 - 4$ 在 $u = \pm \frac{1}{\sqrt{3}}$ 处变号,
故有 2 个拐点. 选 C.

14. 对 a 求导得 $f'(x+a) = \frac{(x+a)[(x+a)^2+1]}{f(x+a)}$, 即 $f(t)f'(t) = t^3 + t$,
$$\frac{f^2}{2} = \frac{t^4}{4} + \frac{t^2}{2} + C.$$

由 $f(1) = \sqrt{2}$ 得 $C = \frac{1}{4}$, 故 $f^2 = \frac{(x^2+1)^2}{2}$,

$$f(x) = \frac{\sqrt{2}}{2}(1+x^2).$$

15. $P\{X \leq 2\} = \frac{1}{2}$, $P\{-1 < X \leq 2\} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

$$P\{\max \leq 2, \min \leq -1\} = P\{\max \leq 2\} - P\{-1 < X \leq 2, -1 < Y \leq 2\} = \frac{1}{4} - \frac{1}{36} = \frac{2}{9}.$$

16. 特征方程 $\lambda^2 - \lambda - 2 = 0$, $\lambda = 2, -1$, $f = C_1e^{2x} + C_2e^{-x}$;

由 $f(0) = 0, f'(0) = 1$ 得 $C_1 = \frac{1}{3}, C_2 = -\frac{1}{3}$:

$$f(x) = \frac{e^{2x} - e^{-x}}{3}, \quad a_n = f^{(n)}(0) = \frac{2^n - (-1)^n}{3}.$$

17. 记 $P_0 = (\alpha_1, \alpha_2, \alpha_3)$, 则 $AP_0 = P_0B$, $B = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 2 \\ 1 & 1 & 3 \end{pmatrix}$.

(I) $|\lambda E - B| = (\lambda - 1)^2(\lambda - 4)$, 故 A 的特征值为 $1, 1, 4$.

(II) B 的特征向量: $\lambda = 4$ 取 $(0, 1, 1)^T$; $\lambda = 1$ 取 $(-1, 1, 0)^T, (-2, 0, 1)^T$. 故

$$P = P_0 \begin{pmatrix} 0 & -1 & -2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = (\alpha_2 + \alpha_3, -\alpha_1 + \alpha_2, -2\alpha_1 + \alpha_3), \quad \Lambda = \text{diag}(4, 1, 1),$$

$$|A - 2E| = (4 - 2)(1 - 2)(1 - 2) = 2.$$

18. Ω 是球心 $(0, 0, 1)$ 、半径 1 的球体. 作平移 $z = 1 + w$, 则

$$m = \iiint_{\Omega} x^2 dV = \iiint_{x^2+y^2+w^2 \leq 1} x^2 dV = \frac{1}{3} \iiint \rho^2 dV = \frac{1}{3} \int_0^1 \rho^2 \cdot 4\pi \rho^2 d\rho = \frac{4\pi}{15}.$$

19. 左端 $= e^{\lim_{x \rightarrow \frac{\pi}{2}} x^2 \ln \cos \frac{\pi}{x}} = e^{-\frac{\pi^2}{2}}$; 右端由积分中值定理 $= \lim_{\xi \rightarrow \infty} f(\xi) = \frac{1}{e}$.

故 $\frac{a^2}{2} = 1$, $a = \sqrt{2}$.

20. 在 $(\frac{1}{2}, \frac{1}{2})$ 处由 $e^z + z = 1$ 得 $z = 0$. 两边求偏导:

$$e^{2yz} \cdot 2yz'_x + 1 + z'_x = 0 \Rightarrow z'_x = -\frac{1}{2}; \quad e^{2yz}(2z + 2yz'_y) + 2y + z'_y = 0 \Rightarrow z'_y = -\frac{1}{2}.$$

故 $dz|_{(\frac{1}{2}, \frac{1}{2})} = -\frac{1}{2}dx - \frac{1}{2}dy$.

21. $A^4 = O$, 故

$$(E + A)^{-1} = E - A + A^2 - A^3 = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

22. 一阶齐次线性方程的解全体为特解的常数倍: $y = C \cos 2x$. 由 $y(0) = 2$ 得 $C = 2$. 选 B.

23. 区域关于 x 轴对称, $\bar{y} = 0$.

$$S = \int_0^{\frac{\pi}{2}} d\theta + \int_{\frac{\pi}{2}}^{\pi} (1 + \cos \theta)^2 d\theta = \frac{5\pi}{4} - 2,$$

$$\iint x d\sigma = \frac{2}{3} \left[1 + \int_{\frac{\pi}{2}}^{\pi} \cos \theta (1 + \cos \theta)^3 d\theta \right] = \frac{5\pi}{8} - \frac{4}{3}.$$

故形心为 $(\frac{15\pi - 32}{6(5\pi - 8)}, 0)$.

24. $xe^x - e^x + 1 = \sum_{n=1}^{\infty} \frac{n-1}{n!} x^n \Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{n+1}{(n+2)!} x^n \quad (-\infty < x < +\infty).$

又

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = \sum_{n=1}^{\infty} \left[\frac{1}{n!} - \frac{1}{(n+1)!} \right] = 1.$$

25. 指数分布无记忆性: $P\{X > 16 \mid X > 8\} = P\{X > 8\} = e^{-8\lambda}$.

26. 在 Σ 上 $x^2 + y^2 + z^2 = 2y$, 故被积函数 $= 2y + y^2 + 2z^2$. 球心 $(0, 1, 0)$ 、半径 1,

$$\iint 2y dS = 2 \cdot 1 \cdot 4\pi = 8\pi, \quad \iint y^2 dS = 4\pi + \frac{4\pi}{3} = \frac{16\pi}{3}, \quad \iint z^2 dS = \frac{4\pi}{3},$$

$$I = 8\pi + \frac{16\pi}{3} + \frac{8\pi}{3} = 16\pi.$$

27. 由题设 $f(1, 1) = 1$, 且 f 在 $(1, 1)$ 处的一阶偏导数与 $e^{x^2+y^2-2}$ 相同: $f'_1 = f'_2 = 2$.

(I) $g'_x = f'_1 e^{x-y} + f'_2 y|_{(1,1)} = 4$, $g'_y = -f'_1 e^{x-y} + f'_2 x|_{(1,1)} = 0$, 故

$$dg|_{(1,1)} = 4 dx.$$

(II)

$$\lim_{t \rightarrow 0} \frac{g(1 + \sin t, 1) - g(1, 1)}{t} - \lim_{t \rightarrow 0} \frac{g(1, 1 - \tan t) - g(1, 1)}{t} = g'_x \cdot 1 - g'_y \cdot (-1) = 4.$$

28. $x = (1 - \cos \theta) \cos \theta$, $y = (1 - \cos \theta) \sin \theta$, 在 $\theta = \frac{\pi}{2}$ 处点为 $(0, 1)$, 且

$$\left. \frac{dx}{d\theta} \right|_{\frac{\pi}{2}} = -1, \quad \left. \frac{dy}{d\theta} \right|_{\frac{\pi}{2}} = 1 \implies k = -1.$$

切线方程: $y - 1 = -x$, 即 $x + y = 1$.

29. (I) 解 $AX = \alpha$ 得

$$X = (1, 1, 0, 0)^T + t(-1, -2, -1, 1)^T.$$

代入 B 的三行分别得 1, 2, -1 (与 t, a 无关), 故这些 X 都满足 $BX = \beta$.

(II) $r(A) = 3$, 解空间维数为 1; 对 B 作行变换 $r_3 + \frac{1}{2}r_2$ 得末行 $(0, 0, a - 1, a - 1)$,

故 $a \neq 1$ 时 $r(B) = 3$, 两方程组解集同为一维, 必同解; $a = 1$ 时 $r(B) = 2$, 解集为二维, 不同解.

故 $a = 1$.

30. 取置换矩阵 $P = (e_2, e_3, e_1) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$, 则 $P^T A P = \text{diag}(2, 3, 1) = B$. 选 B.

31. "矩阵 $(\alpha_1, \dots, \alpha_k)$ 与 $(\beta_1, \dots, \beta_k)$ 等价"即二者秩相同,

而 $r(\alpha_1, \dots, \alpha_k) = k$, 故等价于 $r(\beta_1, \dots, \beta_k) = k$, 即 (II) 线性无关. 选 D.

32. $\frac{\sqrt{n}\bar{X}}{\sigma} \sim N(0, 1)$, $\frac{S_1^2}{\sigma^2} \sim \chi^2(n-1)$ 且二者独立, 故

$$\frac{\sqrt{n}\bar{X}/\sigma}{\sqrt{\frac{S_1^2/\sigma^2}{n-1}}} = \frac{\sqrt{n(n-1)}\bar{X}}{S_1} \sim t(n-1).$$

选 C.

33. $(1+x)^{\frac{2}{x}} = e^{2-x+\frac{2}{3}x^2+o(x^2)} = e^2 \left[1 - x + \frac{7}{6}x^2 + o(x^2) \right], \quad e^2[1 - \ln(1+x)] = e^2 \left[1 - x + \frac{x^2}{2} + o(x^2) \right],$

两者之差 $= \frac{2}{3}e^2x^2 + o(x^2)$, 是关于 x 的高阶无穷小. 选 C.

34. 两边求偏导得 $z'_x = \frac{x-3y}{y+z}$, $z'_y = \frac{-3x+10y-z}{y+z}$.

驻点条件 $x = 3y$, $z = y$, 代入原方程得 $-2y^2 + 18 = 0$, $y = \pm 3$, 故驻点 $(9, 3)$ ($z = 3$) 与 $(-9, -3)$ ($z = -3$) .

在驻点处

$$A = z''_{xx} = \frac{1}{y+z}, \quad B = z''_{xy} = \frac{-3}{y+z}, \quad C = z''_{yy} = \frac{10}{y+z}, \quad AC - B^2 = \frac{1}{(y+z)^2} > 0.$$

故 $(9, 3)$ 处 $A = \frac{1}{6} > 0$, z 取极小值 3; $(-9, -3)$ 处 $A = -\frac{1}{6} < 0$, z 取极大值 -3 .

35. $r(B) \geq 2$ 及 $BA = O$ 给出 $r(A) \leq 3 - 2 = 1$, 故 A 的各行成比例:

$$(a, 1, b) = -(2, -1, 3), \quad (4, c, 6) = 2(2, -1, 3) \implies a = -2, \quad b = -3, \quad c = -2.$$

于是 $A = (1, -1, 2)^T(2, -1, 3)$, $(2, -1, 3)(1, -1, 2)^T = 9$, 故

$$A^n = 9^{n-1}A = 9^{n-1} \begin{pmatrix} 2 & -1 & 3 \\ -2 & 1 & -3 \\ 4 & -2 & 6 \end{pmatrix}.$$

36. $y^* = e^{-x} + xe^x$. 其中 e^{-x} 必为齐次解, xe^x 为对应 ce^x 的特解且出现共振, 说明特征根为 -1 与 1. 故通解

$$y = C_1 e^{-x} + C_2 e^x + xe^x.$$

37.

$$a_n = \int_0^{n\pi} |x \sin x| dx = \sum_{k=1}^n (2k-1)\pi = \pi n^2,$$

$$\sum_{k=1}^{\infty} \frac{1}{\sqrt{a_k a_{k+1}}} = \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \frac{1}{\pi}.$$

选 B .

38. $E(X^2) = a_2$, $D(X^2) = a_4 - a_2^2$, 由中心极限定理

$$Y_n = \frac{1}{n} \sum X_i^2 \text{ 近似服从 } N\left(a_2, \frac{a_4 - a_2^2}{n}\right).$$

选 A .

39. 在 L 上 $x^2 + y^2 = 1$, 故 $e^{x^2+y^2} = e$, 且由对称性 $\int_L xy ds = 0$,

$$\int_L (x-y)^2 e^{x^2+y^2} ds = e \int_L (1-2xy) ds = e \cdot \pi = \pi e.$$

40. 在 I_1 中用 $x \rightarrow \pi - x$: 被积函数中除 x 外关于 $\frac{\pi}{2}$ 对称, 故 $I_1 = \frac{\pi}{2} I_2 > I_2$.

又 $I_2 = 2 \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + e^{\cos^2 x}} dx$ (令 $x \rightarrow \frac{\pi}{2} - t$) $= 2I_3 > I_3$.

故 $I_1 > I_2 > I_3$, 选 A .

41. $A = (a-1)E + J$ (J 为全 1 矩阵), 特征值为 $a-1+n$ (单) 与 $a-1$ ($n-1$ 重) .

(I) 取

$$P = ((1, 1, \dots, 1)^T, (1, -1, 0, \dots, 0)^T, \dots, (1, 0, \dots, 0, -1)^T),$$

则 $P^{-1}AP = \Lambda = \text{diag}(a+n-1, a-1, \dots, a-1)$.

(II) $a \neq 1$ 且 $a \neq 1-n$ 时 A 可逆, $r(A^*) = n$;

$a = 1-n$ 时 $r(A) = n-1$, $r(A^*) = 1$; $a = 1$ 时 $r(A) = 1 \leq n-2$, $r(A^*) = 0$.

42. (I) 令 $g(x) = f(x) - 2(1-x)$, 则 $g(0) = -2 < 0$, $g(1) = 1 > 0$, 由介值定理存在 $x_0 \in (0, 1)$ 使 $f(x_0) = 2(1-x_0)$.

(II) 在 $[0, x_0]$ 、 $[x_0, 1]$ 上分别用拉格朗日中值定理:

$$f'(\xi) = \frac{f(x_0) - f(0)}{x_0 - 0} = \frac{2(1-x_0) + 2}{x_0} = \frac{2(1-x_0)}{x_0}, \quad f'(\eta) = \frac{f(1) - f(x_0)}{1 - x_0} = \frac{1 - 2(1-x_0)}{1 - x_0} = \frac{2x_0 - 1}{1 - x_0},$$

其中 $\xi \in (0, x_0)$, $\eta \in (x_0, 1)$, 故 $\xi \neq \eta$, 且

$$f'(\xi)[1 + f'(\eta)] = \frac{2(1-x_0)}{x_0} \cdot \frac{x_0}{1-x_0} = 2.$$

43. 偏导数存在且连续 $\Rightarrow$ 可微, 与"不可微"矛盾, 故 B 一定不成立. 选 B .

44. 令 $u = 1-x$, $v = 1-y$, 则 $2x - x^2 = 1 - u^2$,

$$I = \iint_{[0,1]^2} \max\{1-u^2, v^2\} du dv, \quad 1-u^2 \geq v^2 \iff u^2 + v^2 \leq 1.$$

记 Q 为四分之一圆盘, 则

$$I = \iint_Q (1-u^2) + \left[\iint_{[0,1]^2} v^2 - \iint_Q v^2 \right] = \left(\frac{\pi}{4} - \frac{\pi}{16} \right) + \left(\frac{1}{3} - \frac{\pi}{16} \right) = \frac{\pi}{8} + \frac{1}{3}.$$

45. 由 $a_{ij} = A_{ij}$ 知 $A^* = A^T$, 故 $AA^T = AA^* = |A|E$, 取行列式得 $|A|^2 = |A|^3$,
 $|A| = 0$ 时由 $AA^T = O$ 得 $A = O$ 与 $a_{33} \neq 0$ 矛盾, 故 $|A| = 1$, A 为正交矩阵.
 而 $b = (a_{13}, a_{23}, a_{33})^T$ 恰为 A 的第三列, 即 $b = A(0, 0, 1)^T$, 故方程组有唯一解
 $x = (0, 0, 1)^T$.

46. 与路径无关 $\iff \frac{\partial}{\partial y}(Fy) = \frac{\partial}{\partial x}(Fx)$, 即 $yF'_y = xF'_x$, 其通解为 $F = \varphi(xy)$.
 故 $F(x, y) = 0$ 确定的曲线为 $xy = C$; 过 $(1, 2)$ 得 $C = 2$, 即

$$xy = 2 \quad \left(y = \frac{2}{x}, \text{ 与坐标轴无交点} \right).$$

47. $P\{X > 1, Y > -1\} = 1 - P\{X \leq 1\} - P\{Y \leq -1\} + P\{X \leq 1, Y \leq -1\},$

而 $P\{X \leq 1\} = \Phi\left(\frac{1}{\sigma}\right)$, $P\{Y \leq -1\} = 1 - \Phi\left(\frac{1}{\sigma}\right)$, 二者之和为 1, 故

$$P\{X > 1, Y > -1\} = \frac{1}{4}.$$

48. 放入的两球中白球数 $W \sim B(2, \frac{3}{5})$, $X = 3 + W$,

$$EX = 3 + \frac{6}{5} = \frac{21}{5}, \quad DX = DW = 2 \cdot \frac{3}{5} \cdot \frac{2}{5} = \frac{12}{25}.$$

选 B.

49. 解方程得 $y = 2e^x - x - 1$, 故

$$y\left(\frac{1}{n}\right) - 1 - \frac{1}{n} = 2\left(e^{\frac{1}{n}} - 1 - \frac{1}{n}\right) = \frac{1}{n^2} + O\left(\frac{1}{n^3}\right),$$

与 $\sum \frac{1}{n^2}$ 同阶, 故级数 (绝对) 收敛.

50. 由曲率公式 $\frac{|y''|}{(1+y'^2)^{3/2}} = \frac{1}{\sqrt{1+y'^2}}$ 及上凸 ($y'' < 0$) 得 $y'' = -(1+y'^2)$.

令 $p = y'$: $\arctan p = -x + C$, 由 $y'(0) = 1$ 得 $C = \frac{\pi}{4}$, 故 $y' = \tan\left(\frac{\pi}{4} - x\right)$,

$$y = \ln \left| \cos\left(\frac{\pi}{4} - x\right) \right| + C_2,$$

由 $y(0) = 1$ 得曲线

$$y = 1 + \ln \left[\sqrt{2} \cos\left(x - \frac{\pi}{4}\right) \right].$$

51. 由 $A^* = 3A^{-1} = -A + 4E$ 得 $A^2 - 4A + 3E = O$, 即 $(A - E)(A - 3E) = O$, 特征值只能为 1, 3;
又 $|A| = 3$, 故特征值为 1, 1, 3, 且 A 可对角化. 于是 $2E - A$ 的特征值为 1, 1, -1, 可逆,
从而 $(2E - A)^T(2E - A)$ 正定, 规范形为 $y_1^2 + y_2^2 + y_3^2$. 选 C.

52.
$$a^{\frac{1}{x}} - a^{\frac{1}{x+1}} = e^{\frac{\ln a}{x}} - e^{\frac{\ln a}{x+1}} \sim \ln a \left(\frac{1}{x} - \frac{1}{x+1} \right) = \frac{\ln a}{x(x+1)} \sim \frac{\ln a}{x^2},$$

故 $x^P \cdot \frac{\ln a}{x^2}$ 有极限当且仅当 $P \leq 2$, 即 $P \in (-\infty, 2]$.

53. $f'_x(x_0, y_0)$ 存在说明 $f(x, y_0)$ 在 x_0 处可导, 从而在该点连续, $\lim_{x \rightarrow x_0} f(x, y_0) = f(x_0, y_0)$ 存在.

A、B 需要更强条件, D 也不必成立 (偏导存在只要求在两条直线上有定义). 选 C.

54. 最大值 $\frac{1}{\sqrt{2\pi}\sigma} = \frac{1}{2\sqrt{2\pi}} \Rightarrow \sigma = 2$, 最大值点 $x = \mu = 1$, 故

$$P\{X < 3\} = \Phi\left(\frac{3-1}{2}\right) = \Phi(1).$$

55. 交线满足 $z^2 = x^2 + y^2 = 2x$, 投影区域 $D: x^2 + y^2 \leq 2x$. 锥面上 $dS = \sqrt{2} dx dy$,

$$\mu = 9\sqrt{2(x^2 + y^2)} = 9\sqrt{2}r,$$

$$m = \iint_D 9\sqrt{2}r \cdot \sqrt{2} d\sigma = 18 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} r^2 dr d\theta = 18 \cdot \frac{32}{9} = 64.$$

56. f 为凹 (下凸) 函数且 $f(0) = 0$, 故对 $t \in [0, 1]$ 有 $f(tx) \leq tf(x)$. 于是

$$\int_0^{\frac{1}{2}} f(x) dx \stackrel{x=u/2}{=} \frac{1}{2} \int_0^1 f\left(\frac{u}{2}\right) du \leq \frac{1}{2} \cdot \frac{1}{2} \int_0^1 f(u) du = \frac{1}{4} \int_0^1 f(x) dx,$$

即 $4 \int_0^{1/2} f \leq \int_0^1 f$. 选 A.

57. (I) 中 $\alpha_1 - 2\alpha_2 + \alpha_3 = 0$, 故向量组 (I) 的秩为 2, 等价要求 β_1, β_2 线性无关且都属于 $\text{span}\{\alpha_1, \alpha_2\}$.

解 $c_1\alpha_1 + c_2\alpha_2 = \beta_1$ 得 $c_1 = -5, c_2 = 6$ (相容); 解 $c_1\alpha_1 + c_2\alpha_2 = \beta_2$ 得 $c_1 = -2, c_2 = 3$, 于是 $a = c_1 + c_2 = 1, b = c_2 = 3$ (此时 β_1, β_2 确实线性无关). 故 $a = 1, b = 3$.

58. (I) $P\{X = k\} = (1-p)^{k-1}p$ ($k = 1, 2, \dots$).

(II) $EX = \frac{1}{p}$, 令 $\bar{X} = \frac{1}{p}$ 得矩估计 $\hat{p} = \frac{1}{\bar{X}}$;

似然函数 $L = p^n(1-p)^{\sum x_i - n}$, $\frac{d \ln L}{dp} = \frac{n}{p} - \frac{\sum x_i - n}{1-p} = 0$ 得最大似然估计

$$\hat{p} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}.$$

59. (I) $\frac{1}{n^2-1} = \frac{1}{2}(\frac{1}{n-1} - \frac{1}{n+1})$, 利用 $\sum_{m \geq 1} \frac{(1/2)^m}{m} = \ln 2$ 得

$$\sum_{n=2}^{\infty} \frac{1}{(n^2-1)2^n} = \frac{1}{2} \left[\frac{\ln 2}{2} - (2 \ln 2 - \frac{5}{4}) \right] = \frac{5}{8} - \frac{3}{4} \ln 2.$$

(II) $\sum_{n=0}^{\infty} \frac{2n+1}{n!} = 2 \sum_{n \geq 1} \frac{1}{(n-1)!} + \sum_{n \geq 0} \frac{1}{n!} = 2e + e = 3e$.

60.
$$d = \frac{|2 \cdot 1 - (-1) + 5 \cdot 2 - 12|}{\sqrt{4+1+25}} = \frac{1}{\sqrt{30}} = \frac{\sqrt{30}}{30}.$$

61. 由 $(q+p)^n = \sum C_n^k p^k q^{n-k}$ 与 $(q-p)^n = \sum C_n^k (-p)^k q^{n-k}$ 相减得

$$P\{A \text{ 发生奇数次}\} = \frac{1 - (1-2p)^n}{2}.$$

62. 记 $s = x + y$. $[1+s] = 1, 2, 3, 4$ 分别对应 $s \in [0, 1), [1, 2), [2, 3), [3, 4)$, 其在正方形内的面积依次为

$$\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{1}{2},$$

故

$$I = 1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{2} + 3 \cdot \frac{3}{2} + 4 \cdot \frac{1}{2} = 10.$$

63. (原题所求量缺失.) 由方程两边对 x 求导:

$$1 = \sin^2 \frac{\pi(y-x)}{4} \cdot (y' - 1) \implies y' = 1 + \csc^2 \frac{\pi(y-x)}{4}.$$

又 $x = 0$ 时 $y - x = 1$, 即 $y(0) = 1$, 此时 $y'(0) = 1 + \frac{1}{\sin^2 \frac{\pi}{4}} = 3$.

64. 由 $A^2 = B^2 = E$ 知 $|A|^2 = |B|^2 = 1$, 结合 $|A| + |B| = 0$ 得 $|A||B| = -1$. 又

$$A(A+B)B = A^2B + AB^2 = B + A = A+B,$$

两边取行列式: $|A||A+B||B| = |A+B|$, 即 $-|A+B| = |A+B|$, 故 $|A+B| = 0$, $A+B$ 不可逆.

65. 取 $x = y = 0$ 得 $f(0) = 0$; 再由定义

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^x f(h) + e^h f(x) - f(x)}{h} = e^x f'(0) + f(x) = e^{x+1} + f(x).$$

解线性方程 $f' - f = e \cdot e^x : (fe^{-x})' = e$, $f = (ex + C)e^x$, 由 $f(0) = 0$ 得 $C = 0$:

$$f(x) = xe^{x+1}.$$

66. 由 $u = x + y$, $v = x - y$ 得 $\partial_x = \partial_u + \partial_v$, $\partial_y = \partial_u - \partial_v$, 从而

$$z_{xx} + 2z_{xy} + z_{yy} = 4 \frac{\partial^2 z}{\partial u^2} = 0.$$

又 $xy = \frac{u^2 - v^2}{4}$, $w = \frac{u^2 - v^2}{4} - z$, 故

$$(I) \quad \frac{\partial^2 w}{\partial u^2} = \frac{1}{2}.$$

(II) 积分得 $\frac{\partial w}{\partial u} = \frac{u}{2} + \varphi(v)$, 由 $\frac{\partial w(0,v)}{\partial u} = ve^{-v}$ 得 $\varphi(v) = ve^{-v}$;

再积分 $w = \frac{u^2}{4} + uve^{-v} + \psi(v)$, 由 $w(0,v) = \frac{v^2}{4}$ 得 $\psi(v) = \frac{v^2}{4}$. 于是

$$z = xy - w = \frac{u^2 - v^2}{4} - \frac{u^2}{4} - uve^{-v} - \frac{v^2}{4} = -\frac{v^2}{2} - uve^{-v},$$

即

$$z(x, y) = -\frac{(x - y)^2}{2} - (x + y)(x - y)e^{-(x - y)}.$$

67. 被积表达式 $= d\left[\frac{1}{2} \ln(x^2 + y^2)\right] - \frac{x dy - y dx}{x^2 + y^2}$.

补上从 $B(-\pi, -\pi)$ 到 $A(\pi, -\pi)$ 的直线段, 所得闭曲线为逆时针且围住原点, 故

$$\oint = 0 - 2\pi = -2\pi.$$

而在该直线段上 ($y = -\pi$, $dy = 0$)

$$\int_{-\pi}^{\pi} \frac{x - \pi}{x^2 + \pi^2} dx = 0 - \pi \cdot \frac{1}{\pi} \left[\arctan \frac{x}{\pi} \right]_{-\pi}^{\pi} = -\frac{\pi}{2}.$$

$$\text{故 } I = -2\pi - \left(-\frac{\pi}{2}\right) = -\frac{3\pi}{2}.$$

68. 由 $a_{ij} = A_{ij}$ 得 $A^* = A^T$, 故 $AA^T = |A|E = E$, A 为正交矩阵.

于是第三行为单位向量: $a_{31}^2 + a_{32}^2 + a_{33}^2 = 1$, 而 $a_{33} = 1$, 故 $a_{31} = a_{32} = 0$.

由克拉默法则 $x_i = \frac{A_{3i}}{|A|} = a_{3i}$, 故解为

$$x = (0, 0, 1)^T.$$

69. A 中 $k = 0$ 时不是特征向量; B 、 C 中 $A(\alpha_1 \pm \alpha_2) = \alpha_1 \mp \alpha_2$ 与 $\alpha_1 \pm \alpha_2$ 不成比例 (α_1, α_2 线性无关).

$D: A^2(\alpha_1 + \alpha_2) = \alpha_1 + \alpha_2$, 且 $\alpha_1 + \alpha_2 \neq 0$, 故是 A^2 的特征向量. 选 D .

70. (I) 逐项求导：

$$y' = \sum_{n \geq 1} \frac{x^{3n-1}}{(3n-1)!}, \quad y'' = \sum_{n \geq 1} \frac{x^{3n-2}}{(3n-2)!},$$

三者相加恰好取遍所有幂次，故 $y'' + y' + y = \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$.

(II) 由 $y(0) = 1, y'(0) = 0$ 解该方程：特征根 $\frac{-1 \pm \sqrt{3}i}{2}$ ，特解 $\frac{1}{3}e^x$ ，

$$y = e^{-\frac{x}{2}} \left(C_1 \cos \frac{\sqrt{3}}{2}x + C_2 \sin \frac{\sqrt{3}}{2}x \right) + \frac{e^x}{3}, \quad C_1 = \frac{2}{3}, C_2 = 0,$$

故

$$\sum_{n=0}^{\infty} \frac{x^{3n}}{(3n)!} = \frac{2}{3}e^{-\frac{x}{2}} \cos \frac{\sqrt{3}}{2}x + \frac{e^x}{3}.$$

71. Y 的密度为 $f(-y)$ 说明 Y 与 $-X$ 同分布，故 $DY = DX = 1$ ， $\text{Cov}(X, Y) = \rho\sqrt{DXDY} = -\frac{1}{2}$ ，
 $D(X - Y) = DX + DY - 2\text{Cov}(X, Y) = 1 + 1 + 1 = 3$.

选 D .

72. $x \rightarrow 0$ 时 $|f(x)| = \left| \frac{(x^3-1)\sin x}{|x|(1+x^2)} \right| \rightarrow 1$ ，有界；
 $|x| \rightarrow \infty$ 时 $\left| \frac{x^3-1}{|x|(1+x^2)} \right| \rightarrow 1$ ，而 $|\sin x| \leq 1$ ，也有界。

故 f 在 $(-\infty, +\infty)$ 内有界。选 A .

73. 齐次型：令 $u = \frac{y}{x}$ ， $xu' = \tan u$ ， $\cot u du = \frac{dx}{x}$ ，得 $\sin u = Cx$ ，即 $\sin \frac{y}{x} = Cx$ 。
 由 $y(1) = \frac{\pi}{6}$ 得 $C = \frac{1}{2}$ ，故特解

$$\sin \frac{y}{x} = \frac{x}{2}, \quad \text{即} \quad y = x \arcsin \frac{x}{2}.$$

74. 配方：

$$f = x_1^2 - x_3^2 + 4x_1x_2 + 4x_2x_3 = (x_1 + 2x_2)^2 - (2x_2 - x_3)^2.$$

令 $y_1 = x_1 + 2x_2, y_2 = 2x_2 - x_3, y_3 = x_3$ ，反解得

$$x_1 = y_1 - y_2 - y_3, \quad x_2 = \frac{y_2 + y_3}{2}, \quad x_3 = y_3,$$

即 $C = \begin{pmatrix} 1 & -1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}$. 选 A .

75. 法向量 $(2x, 4y, 6z) \parallel (1, 4, 6)$ 给出 $y = 2x, z = 2x$ ；代入曲面方程得 $21x^2 = 21, x = \pm 1$ ，
 切点为 $(1, 2, 2)$ 与 $(-1, -2, -2)$ ，相应切平面为

$$x + 4y + 6z = 21 \quad \text{与} \quad x + 4y + 6z = -21.$$